Experimental Stress Analysis

UNIT – I

PRINCIPLES OF EXPERIMENTAL APPROACH

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Merits of Experimental Analysis – Introduction, Uses of Experimental Stress Analysis – Advantages of Experimental Stress Analysis – Different Methods – Simplification of Problems.

1. Introduction to Experimental Stress Analysis (ESA)

- Experimental Stress Analysis = the branch of solid mechanics in which stresses, strains and deformations in a structure or machine part are found by measurement on the real part (prototype) or on a scale model, instead of (or along with) calculation.
- Stress cannot be measured directly. We always measure a strain, a displacement, a force or an optical effect and then convert it to stress using the material's elastic constants (E, ν, G).

$$\sigma = E \cdot \varepsilon \text{ (uniaxial)} \quad \sigma = (E / (1 - \nu^2))(\varepsilon_x + \nu \varepsilon_y) \text{ (biaxial)}$$

- The aim of design: every point of the component should be safe, economical and reliable. For that the designer must know the magnitude, location and direction of the maximum stress.

Why do we need experiments?

- Theory of elasticity gives exact answers only for simple shapes and simple loads (beams, plates, cylinders, discs).
- Real structures have holes, notches, fillets, threads, bolted joints, cut-outs – stress concentration zones where formulae fail.
- Material may be non-homogeneous, anisotropic or non-linear (concrete, composites, soils, rock).
- Boundary conditions, load transfer and support behaviour are uncertain – experiments give the truth, theory gives only an estimate.
- Codes and standards themselves are built on experimental evidence (e.g. cube tests, beam tests).

Brief history (remember 4–5 landmarks)

- 1816 Brewster – discovered photoelasticity (temporary double refraction).
- 1856 Lord Kelvin – showed resistance of a wire changes with strain (basis of strain gauge).
- 1930s Coker and Filon – 2-D photoelasticity became practical.
- 1938 Simmons and Ruge – first bonded wire resistance strain gauge.
- 1950s Foil gauges (printed circuit technique) and brittle-lacquer coatings.

- 1960s onwards – moiré, holography, speckle interferometry, and now digital image correlation (DIC) and fibre-optic sensors.

Background revision (needed for every unit)

- Normal strain $\varepsilon = \Delta L / L$ (change in length per unit length; dimensionless, often quoted in micro-strain, $1\ \mu\varepsilon = 10^{-6}$).
- Shear strain γ = change in the right angle between two lines (radians); Poisson's ratio ν = – lateral strain / axial strain.

$$E = 2G(1 + \nu)\quad K = E / [3(1 - 2\nu)]\quad G = E / [2(1 + \nu)]$$

- Principal stresses (planes of zero shear) for a 2-D state:

$$\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{[\,((\sigma_x - \sigma_y)/2)^2 + \tau_{xy}^2\,]}$$
$$\tan 2\theta = 2\tau_{xy} / (\sigma_x - \sigma_y)\quad \tau_{max} = (\sigma_1 - \sigma_2)/2$$

- Same equations hold for strain with $\sigma \to \varepsilon$ and $\tau \to \gamma/2$ (used in rosette analysis, Unit III).

2. Merits of Experimental Analysis

- Direct and real: result is obtained on the actual material and the actual boundary conditions – no idealisation.
- Works for complex geometry where the mathematical solution is impossible or too lengthy.
- Gives the full-field picture (photoelasticity, brittle coating, moiré) – one look shows the critical zone.
- Checks theory: an experimental result is the standard against which analytical and FEM results are validated.
- Material independent: applicable to metals, plastics, concrete, rock, bone, composites.
- Static as well as dynamic loads (impact, vibration, fatigue, thermal) can be studied.
- Simple models can be tested; small model results are scaled up to prototype by similitude.
- Finds unknown quantities – residual stresses, thermal stresses, contact stresses, assembly stresses – which no designer can calculate easily.

- Safe and economical – failure of a model costs much less than failure of a dam or bridge.
- Builds confidence – used by the designer, the contractor and the court (failure investigation).

Limitations (examiners also ask!)

- Needs skilled operators, costly instruments and a laboratory.
- Each test gives the answer only for that one structure and that loading; parametric studies are slow.
- Model material and scaling errors; gauge installation errors.
- Result is point-wise (strain gauges) or surface only (most optical methods).

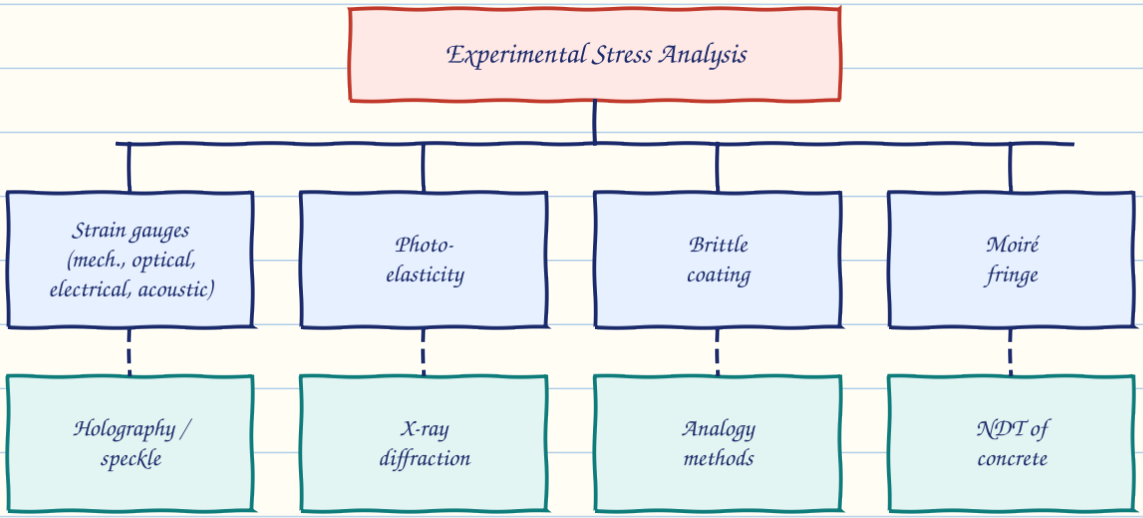


Fig 1.1 Classification of experimental stress analysis methods

3. Uses of Experimental Stress Analysis

- Design verification – proof testing of bridges, dams, buildings, pressure vessels, aircraft and machine frames.
- Finding stress concentration factor (K_t) at holes, notches, keyways, fillets.

$$K_t = \sigma_{max} / \sigma_{nominal}$$

- Locating the critical points so that strain gauges / sensors can be fixed in the right place; also for structural health monitoring.
- Model testing of large structures – dams, shell roofs, chimneys, bridges – before construction (e.g. dam models in laboratories).

- *Residual stress measurement* – welds, castings, rolled sections, shrink fits (hole drilling, X-ray).
- *Measurement of load, torque, pressure, force* (load cells, proving rings, torque meters, pressure transducers use strain gauges!).
- *Material characterisation* – determination of E, ν, ultimate strength, fatigue strength.
- *Failure analysis* – finding why a component cracked or collapsed.
- *Quality control and NDT* – checking concrete strength, cracks, voids (rebound hammer, UPV).
- *Research* – developing new materials, new analytical and numerical methods; validating FEM.

Examples from civil engineering

- Strain gauges on *steel bridge girders* during a load test – compare measured and calculated stresses.
- *Photoelastic model* of a dam / tunnel lining / pile cap to see stress contours.
- *Rebound hammer and UPV* on an old RCC building to assess quality and strength.
- *Vibrating wire gauges* embedded in concrete dams to monitor temperature strains.

4. Advantages of Experimental Stress Analysis

- No need for any *assumptions* about stress distribution – the structure itself tells you.
- *Full-field methods* give stress at every point; *point methods* give accurate numerical values.
- *High accuracy* – modern gauges measure to 1 $\mu\varepsilon$ (10^{-6} strain) or better; error about 1–2 %.
- Usable on *very small* (microscopic) and *very large* (dam) structures.
- *In-service monitoring* is possible – structure is not destroyed (non-destructive).
- Results are *recorded electrically* – easy storage, remote reading, data loggers, computer processing.
- *Fast* for complicated shapes: a photoelastic model can be cut in a few hours.

- Complements numerical methods (FEM, BEM) – both together give a reliable design.

★ Remember: Experimental stress analysis does NOT replace theory – it checks and extends it.

Comparison table (write in answer)

- Analytical method: exact for simple cases only; cheap; fast; many assumptions.
- Numerical (FEM): handles complex geometry; depends on mesh and boundary conditions; needs validation.
- Experimental: real material and real loading; shows true behaviour; needs laboratory and skill; costly for each case.

5. Different Methods of Experimental Stress Analysis

5.1 Strain-gauge methods (Units II and III)

- Measure the change of length / resistance / inductance / frequency over a small gauge length.
- Mechanical (Huggenberger, Berry, dial gauge), optical (Martens mirror, Tuckerman), acoustic (vibrating wire), electrical (inductance, LVDT, capacitance, resistance gauges – most popular).
- Gives strain at a point on the surface; stress from Hooke's law; rosettes give principal strains.

5.2 Photoelasticity (Units IV and V)

- Transparent model of birefringent plastic (epoxy, polycarbonate) is loaded and viewed in polarised light.
- Fringe patterns (isochromatics and isoclinics) give the principal stress difference and direction over the whole model.
- Two-dimensional (transmission), three-dimensional (stress freezing), reflection photoelasticity (birefringent coating on real part).

5.3 Brittle coating method

- Surface of the part is coated with a brittle lacquer (Stresscoat). When the part is loaded, the coat cracks perpendicular to maximum principal tensile strain.

- Crack pattern shows location and direction of critical stress; quick, qualitative; strain level estimated from calibration strip.

5.4 Moiré fringe method

- Two gratings (specimen grating and reference grating) superimposed → moiré fringes that map displacement in the in-plane direction.
- Strain = derivative of displacement: $\varepsilon = (1/p)(\partial N/\partial x)$, p = pitch of grating.

5.5 Holography and speckle interferometry

- Laser light, interference between two states of an object gives displacement to a fraction of the wavelength; good for out-of-plane displacement, vibration, crack detection.

5.6 X-ray diffraction

- Measures change in lattice spacing of metals → residual and applied stresses without touching the part.

5.7 Analogy methods

- Equations of stress problems are similar to those of other phenomena – membrane (soap-film) analogy for torsion of non-circular sections, electrical analogy for plane problems, sand-heap analogy for plastic torsion.

5.8 Non-destructive testing (NDT)

- Rebound hammer, ultrasonic pulse velocity (UPV), radiography, dye penetrant – estimate strength, cracks, voids without damage (Unit III).

5.9 Other modern methods

- Digital Image Correlation (DIC) – camera tracks speckle pattern; gives full-field displacement and strain.
- Fibre-optic (FBG) sensors, thermo-elastic stress analysis (SPATE), acoustic emission.

6. Simplification of Problems

- Real structures are three-dimensional; solving them directly is difficult. The first job of an experimenter is to simplify the problem without losing the essential physics.

6.1 Dimensional reduction

3-D body → 2-D (plane stress / plane strain) → 1-D (bar / beam)

- *Plane stress: thin plate loaded in its own plane.* $\sigma_z = \tau_{xz} = \tau_{yz} = 0$. *Used in 2-D photoelastic models (plate with hole, gear tooth, beam).*
- *Plane strain: very long body with uniform section and load (dam, tunnel, retaining wall, culvert).* $\varepsilon_z = \gamma_{xz} = \gamma_{yz} = 0$ *and* $\sigma_z = \nu(\sigma_x + \sigma_y)$.

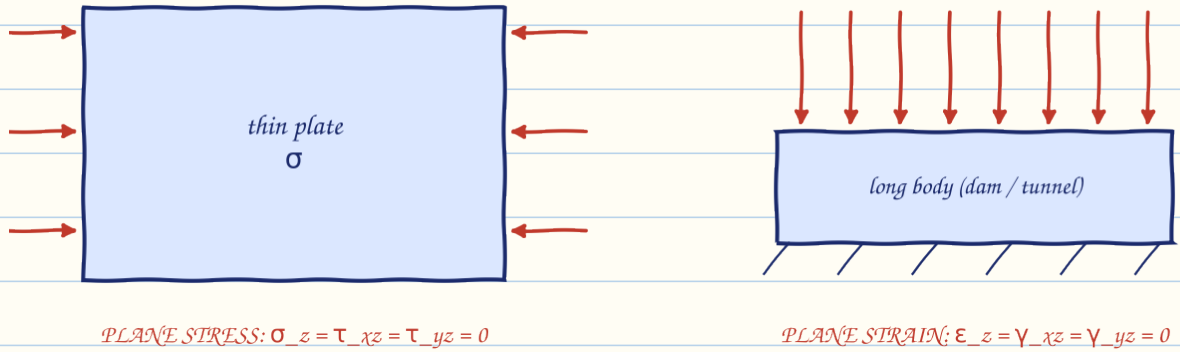


Fig 1.2 Plane stress and plane strain idealisation

6.2 Other simplifications

- *Symmetry: test only one half or one quarter of a symmetric structure with proper boundary conditions.*
- *Saint-Venant's principle: the local effect of a load is felt only up to a distance about equal to the largest dimension of the loaded area; away from it, stresses depend only on the resultant force and moment. So the way a load is applied can be simplified.*
- *Linear elasticity and superposition: if stresses are below the elastic limit, results of separate load cases can be added.*
- *Reduction to a model: use a small-scale, cheaper material model (acrylic, epoxy, micro-concrete) – needs similitude.*
- *Choosing critical sections – measure only where maximum stress is expected (stress concentration zones) rather than everywhere.*
- *Neglect secondary effects (self-weight of small model, friction at supports) if they are small.*

7. Model Theory and Similitude (support for simplification)

- Model and prototype should be geometrically (same shape, all lengths scaled by same factor), kinematically and dynamically similar.

$$\text{Length scale } \lambda_L = L_m / L_p \quad \text{Force scale } \lambda_P = P_m / P_p$$

- For static elastic problems with same Poisson's ratio:

$$\text{Stress scale } \sigma_m / \sigma_n = \lambda_P / \lambda_L^2$$
$$\text{Strain scale } \varepsilon_m / \varepsilon_n = \lambda_P / (\lambda_E \lambda_L^2)$$
$$\text{Deflection scale } \delta_m / \delta_p = \lambda_P / (\lambda_E \lambda_L)$$

- If the model is made of the same material ($\lambda_E = 1$) and loaded so that $\lambda_P = \lambda_L^2$, stresses and strains are equal in model and prototype.
- Buckingham π-theorem is used to form dimensionless groups (e.g. σ/E, $P/(E L^2)$, ν, δ/L) – the groups must be equal for model and prototype.

Example

- A model of a bridge pier is made 1/20 scale, same material. Prototype load = 4000 kN. To get the same stresses, model load $P_m = P_p \lambda_L^2 = 4000 \times (1/20)^2 = 10$ kN.

8. Requirements of a Good Measuring System

- Sensitivity – large output for small strain.
- Range – covers expected strain (elastic strains are only about 10^{-6} to 3×10^{-3}).
- Accuracy and repeatability; linearity and low hysteresis.
- Gauge length short enough to measure local strain (at stress concentration) but long enough to average heterogeneous material (concrete needs long gauges).
- Quick response (frequency response) for dynamic loads.
- Stability – no zero drift; insensitive to temperature, humidity, vibration.
- Light weight, small size (must not stiffen the member), cheap and easy to install.
- Remote reading and recording facilities.

Accuracy, precision, resolution

- Accuracy – closeness of the reading to the true value; precision – closeness of repeated readings to each other; resolution – smallest change the instrument can show (e.g. 1 $\mu\varepsilon$).
- Sensitivity = output change / input change (e.g. mm of pointer travel per micro-strain).

- Types of error: gross (reading/recording mistake), systematic (calibration, zero shift, temperature – can be corrected), random (cannot be avoided – reduced by repeating and averaging).

Sources of error

- Calibration error, gauge-installation error (bad bonding), temperature effect, lead-wire resistance, transverse sensitivity, reinforcing effect of the gauge, human reading error.
- Use statistical treatment: mean $\bar{x} = \Sigma x/n$ and standard deviation $s = \sqrt{[\Sigma (x - \bar{x})^2/(n-1)]}$ to judge the reliability of repeated readings.

9. Steps in an Experimental Investigation

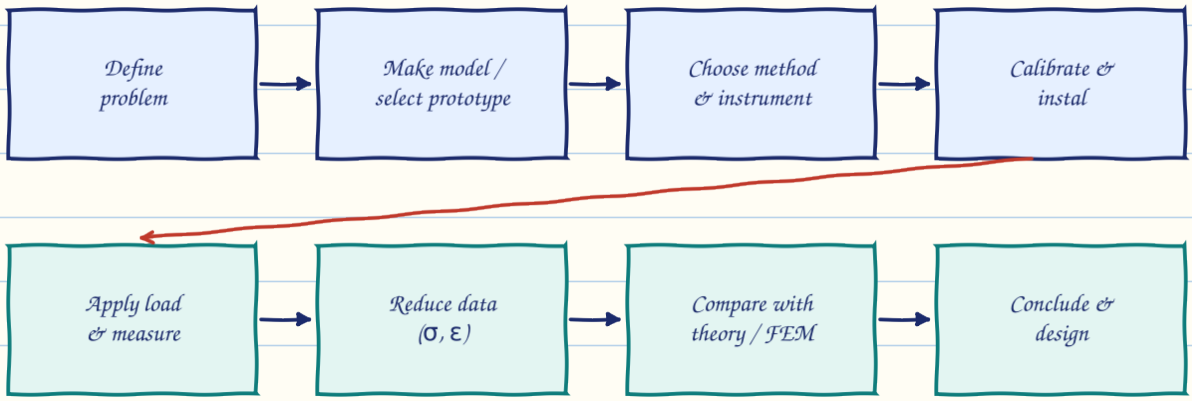


Fig 1.3 Flow chart of an experimental stress analysis

- Define the problem – what quantity ($\sigma, \varepsilon, \delta$) at which location?
- Select the method by considering accuracy, cost, full-field or point, static or dynamic, model or prototype.
- Calibrate the instrument against a known standard.
- Apply loads slowly in steps and record the readings; repeat for reliability.
- Reduce the data – convert to stress using E and ν; plot.
- Compare with theory or FEM, explain differences, give conclusions and design recommendations.

10. Static, Dynamic and Field Load Tests

Static and dynamic measurement

- Static test – load applied slowly; any slow instrument (dial gauge, mechanical gauge, resistance gauge with indicator) can be used.
- Dynamic test – impact, vibration, moving vehicles, earthquakes; needs high frequency response – resistance gauges + oscilloscope / data logger, piezo sensors.
- Gauge used for dynamic work must have natural frequency at least 5–10 times the frequency of the phenomenon.

Procedure of a field load test on a structure (e.g. bridge girder or slab)

- Study the drawings, calculate expected stresses and deflections, decide gauge positions (mid-span, quarter span, near supports).
- Fix strain gauges (bottom and top fibre) and dial gauges / LVDTs for deflection; connect to the data logger and record the zero reading.
- Apply load in increments (25 %, 50 %, 75 %, 100 % of the test load); hold each increment and note readings.
- Hold the full load for the specified time (usually 24 h for a slab, as per IS 456 Cl. 17.6); then unload in stages and check recovery of deflection.
- Compare measured and calculated values; recovery of at least 75 % of the maximum deflection indicates satisfactory behaviour of an RCC member.

★ Linear load–deflection graph and good recovery show that the structure behaved elastically and safely.

11. Solved Numerical Examples

Example 1 – Stress concentration factor

- In a plate with a circular hole the nominal stress is 40 MPa. A gauge at the edge of the hole records a maximum stress of 120 MPa.

$$K_t = \sigma_{max} / \sigma_{nom} = 120 / 40 = 3.0$$

- This agrees with the theoretical value $K_t = 3$ for a small hole in a wide plate (Kirsch solution).

Example 2 – Stress from measured strains (biaxial)

- Steel plate: $E = 200$ GPa, $\nu = 0.3$. Gauges give $\varepsilon_x = 600\ \mu\varepsilon$ and $\varepsilon_y = -100\ \mu\varepsilon$.

$$\sigma_x = E/(1-\nu^2)\,(\varepsilon_x + \nu\,\varepsilon_y) = (200\times10^3/0.91)(600 - 30)\times10^{-6} = 125.3\ \text{MPa}$$

$$\sigma_y = E/(1-\nu^2)\,(\varepsilon_y + \nu\,\varepsilon_x) = (200\times10^3/0.91)(-100 + 180)\times10^{-6} = 17.6\ \text{MPa}$$

- Note: even though ε_y is negative, σ_y is tensile – this is why one gauge alone cannot give stress in a biaxial field.

Example 3 – Model loading

- A 1:10 scale model of a bracket, made of the same material, is to reproduce the prototype stresses. Prototype load = 50 kN.

$$\lambda_L = 1/10, \lambda_E = 1 \rightarrow \lambda_P = \lambda_L^2 = 1/100 \rightarrow P_m = 50/100 = 0.5\ kN$$

- The model deflection is then $\delta_m/\delta_p = \lambda_P/\lambda_L = (1/100)/(1/10) = 0.1$, i.e. 1/10 of prototype deflection.

Example 4 – Plane strain stress

- In a long retaining wall (plane strain) $\sigma_x = 30$ MPa, $\sigma_y = 10$ MPa, $\nu = 0.25$.

$$\sigma_z = \nu(\sigma_x + \sigma_y) = 0.25 \times 40 = 10\ MPa$$

- Although $\varepsilon_z = 0$, a stress σ_z of 10 MPa still exists. In plane stress the reverse is true: $\sigma_z = 0$ but $\varepsilon_z \neq 0$ ($\varepsilon_z = -\nu(\sigma_x + \sigma_y)/E$).

12. Quick Revision – Unit I

- ESA = finding stress and strain by measurement, not calculation.
- Merits: realistic, full-field, handles complex shapes, validates theory.
- Uses: design check, K_t, model tests, residual stress, NDT, failure analysis.
- Methods: strain gauges, photoelasticity, brittle coating, moiré, holography, X-ray, analogies, NDT.
- Simplification: 3-D → 2-D (plane stress / plane strain) → 1-D; symmetry; Saint-Venant; similitude.
- $\sigma_m/\sigma_p = \lambda_P/\lambda_L^2$.

Important questions

- Explain the merits and limitations of experimental stress analysis.
- Write the uses and advantages of ESA in civil engineering.
- Classify the methods of experimental stress analysis and describe any three.
- How are three-dimensional problems simplified? Explain plane stress and plane strain.
- Explain the requirements of a good strain-measuring system.

- State the principle of similitude and derive the stress scale between model and prototype.

References

- Sadhu Singh – Experimental Stress Analysis, Khanna Publishers.
- Dally and Riley – Experimental Stress Analysis, McGraw-Hill.
- L.S. Srinath et al. – Experimental Stress Analysis, Tata McGraw-Hill.